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The title

Fernando Portela

Abstract.

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Contact information:
F. Portela: Independent Researcher. math.nt@elnando.anonaddy.me

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1 Introduction

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2 first section

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3 Conclusion

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Appendix: Dependencies between theorems

The following directed graphs map the logical dependencies between the central formalized results. To maintain structural clarity and avoid overwhelming edge density, basic definitions and foundational types are generally excluded from the hierarchy.

- **Blue-shaded rounded rectangles**: Formal proofs (Lemmas and Theorems).
- **Green-shaded rectangles**: Central definitions or structures.
- **Directed arrows** ($A \to B$): Logical dependence, signifying that proof A explicitly "uses" or "applies" result B to close a goal.

theorem1.png

Figure 1: .